

Leo Zhang

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PROFESSIONAL SUMMARY

Master of Computer and Information Sciences (**First Class Honours**) from AUT. Software Developer — Full-stack engineer with **10+ years** shipping production systems using modern technologies — from Vue.js front-ends to **Java**, C#/.NET and back-end **services**. Strong focus on software **architecture**, SOLID principles and design patterns — building scalable distributed systems that balance quality with business delivery constraints. Delivered end-to-end features across web, mobile and IoT, with hands-on CI/CD and automated testing at scale. Published AI research at IVCNZ 2025, with practical experience integrating AI-assisted development tools and LLM workflows into delivery pipelines.

CORE COMPETENCIES

Languages: C#/.NET, **Java**, **TypeScript**, JavaScript

Frontend: Vue.js, Angular, HTML/CSS

Backend & APIs: Spring Boot, Spring **Cloud**, RESTful API, **Microservices**

Architecture & Practices: SOLID Principles, Design Patterns, Distributed Systems, Unit Testing

Databases & Storage: MySQL, SQL Server, Redis, MongoDB

Cloud & DevOps: Docker, Kubernetes, Azure, **AWS**

EXPERIENCE

Auckland University of Technology — *Auckland, New Zealand*

2024 – 2026

AI Research Engineer

- Designed and delivered ChatClothes, a digital multimodal AI platform integrating computer vision, diffusion models, and LLM-powered workflows with RAG, vector search, and prompt grounding. Conducted extensive image dataset work, model training and testing across multiple CV projects. Provided technical leadership and owned the product end-to-end as **cloud-native AI solutions**, from system design and development through evaluation, deployment, and user validation.
- Deployed and optimized AI models on edge devices including Raspberry Pi and Android phones, running lightweight LLMs and vision models on resource-constrained ARM hardware with offline inference, image processing, and on-device prediction. Validated model outputs through ground truth testing and evaluation.

Tech: Python, PyTorch, Ollama, Docker, Linux, AWS, Azure, Git, Raspberry Pi, Android

Chunxiao Technology Co., Ltd. — *China*

2013 – 2024

*Technology company (est. 2013, ~40 employees) delivering software, hardware, and system integration across streaming media, **enterprise platforms**, IoT, and industrial **automation**.*

Senior Software Engineer | 2018 - 2024

- Smart Factory IoT Platform: architected manufacturing platform across 10+ factory sites with 6-person cross-functional team. Integrated RFID readers, barcode scanners, electronic scales (RS232/RS485 serial bridge to WebSocket), conveyors, and Android workshop terminals with Spring **Cloud microservices**. Event-driven **architecture** via ActiveMQ/Kafka, Docker deployment, 30%+ efficiency gain.
- IoT Device Management Platform: built Android device discovery/binding/remote control app and gateway firmware for smart switches and Zigbee/WiFi gateways. Protocol handling, MQTT **cloud**

uplink, Spring **Cloud** backend API, and Vue.js management dashboard.

- Boobit Crypto Trading App: sole Android developer for financial cryptocurrency trading app. Real-time market data via WebSocket, order book rendering, candlestick charting, encrypted local wallet storage, MVVM with Clean **Architecture**, Retrofit/OkHttp, Room — delivering banking-grade reliability for sensitive financial transactions.
- Picture Book Locker: Android cabinet control system with electromagnetic locks, UV sterilization, sensors, face recognition + QR **code** authentication. Borrowing client app with <3s checkout time, 24/7 self-service.

Tech: Kotlin, Java, Android SDK, CI/CD, WebSocket, NDK/JNI, Spring Cloud, REST APIs, MQTT, Zigbee, RFID

Senior Android Engineer | 2013 - 2018

- **Enterprise** Messaging Platform: architected Android client for 5,000+ DAU and 500K+ daily messages with sub-200ms latency. Built NDK/JNI TCP/UDP transport layer, WebSocket connection persistence, offline message sync, JPush backup push channel, and OEM ROM keep-alive strategies (Xiaomi/Huawei/OPPO). Migrated core IM from unstable in-house C++ to Easemob **cloud**, eliminating 90%+ defects. Built FastDFS distributed file storage (100GB+). Applied hotfix framework (Tinker, AndFix) for emergency patches and performance quality engineering across cold start, ANR, OOM, and APK size.
- Live Streaming Commerce System: full-stack Android + web delivery for live commerce platform. Android client in **Java**/Kotlin with RTMP/WebRTC/HLS streaming, multi-language **architecture** (C# + C++ media core + Python analytics + Lua config), 1000+ peak concurrency, 99.5% stream availability.

Tech: Kotlin, Java, Android SDK, Gradle, CI/CD, WebSocket, NDK/JNI, C++, TCP/UDP, Tinker, AndFix, JPush, RTMP, WebRTC, HLS, Ea

Additional Experience — *Auckland, New Zealand*

Customer Service & Operations (Part-time)

- Gilmours & Red Cross Shop: customer service, inventory management, and team collaboration

EDUCATION

Master of Computer and Information Sciences *Auckland University of Technology, NZ* 2024 – 2026

Research: Computer Vision, Diffusion Models, Large Language Models, Multimodal AI

Bachelor of Software Engineering *Hebei Univ. of Sci. & Tech., China* 2009 – 2013

Coursework: Data Structures, Algorithms, **Java** Programming, Database Systems, Software Engineering

LICENSES & CERTIFICATIONS

IoT Fundamentals — Xi'an Jiaotong University 2020

Science and Technology Achievement Award (Smart Manufacturing) — Hebei, China 2020

Software Design Engineer Certificate — China Qualification Authority 2019

INTERESTS

Hiking, self-hosted **infrastructure**, AI tools, and mobile technology trends

REFEREES

References available upon request.